NETWORK DESIGN PROJECT REPORT FOR GROUP 1

Group 1

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Introduction

The company has moved to a new office with no existing network structure. This project's objective was to design and configure a budget-friendly, expandable, and secure network by using the Cisco Packet Tracer simulation tool. The objectives were:

- to offer wired and wireless connections at each department without interruption
- to offer segregated departments while offering access to printers and the internet
- to create a design to expand in the future and allow for remote connection
- to control cost while offering a stable network

Possible problems were group working and coordination, misconfiguration errors, and VLAN mismatch between sections, and balancing cost with performance of the network, all issues solved by extensive testing, defining roles clearly, and a contingency plan.

Literature Review

The network was designed using a hierarchical model consisting of access, distribution, and core layers. This structure is widely used in enterprise networks because it improves scalability, simplifies troubleshooting, and allows clear separation of responsibilities.

Topology Discussion: The topology follows a **hierarchical star design**, with the **machine room** acting as the central hub. All major routing and switching functions are concentrated in the machine room, ensuring that critical infrastructure is secure and centrally managed. From this hub, connections extend outward to departmental access switches and wireless access points.

A key design requirement was that **no switches with more than 8 ports may be placed outside the machine room**. To comply, all 9–24 port ranges were disabled on switches deployed in offices, reception, kitchen, and the meeting room. This ensures that only compact access switches are used in user areas, while larger multi-port switches remain confined to the machine room. This design choice improves security, reduces the risk of tampering, and simplifies maintenance by limiting high-capacity hardware to a controlled environment.

Access Layer

- Each office, reception, and meeting room is equipped with a small access switch (≤8 ports enabled).
- These switches connect end-user devices such as PCs, printers, and IoT devices.
- Wireless access points are connected to these switches but configured only as APs (DHCP disabled) to avoid routing conflicts.

Distribution Layer

- Traffic from access switches is aggregated into distribution switches located in the machine room.
- VLAN segmentation is applied at this layer to isolate departments and enforce security policies.

Core Layer

- A central router in the machine room performs inter-VLAN routing, DHCP, NAT, and ACL enforcement.
- This router ensures that while departments remain isolated, shared resources (printers, internet) are accessible.

Design Rationale

- **Security:** Limiting large switches to the machine room reduces exposure of critical hardware.
- **Scalability:** The hierarchical model allows easy expansion by adding more access switches without redesigning the core.

- **Maintainability:** Faults are isolated to individual offices or sections, simplifying troubleshooting.
- **Compliance:** Disabling 9–24 ports outside the machine room directly meets the project requirement.

This topology balances robustness, affordability, and compliance with the given constraints, while aligning with modern enterprise networking principles.

Methodology

The network was implemented using cost-effective, scalable hardware to balance performance and budgetary constraints.

A single central router was used instead of a Layer 3 switch. While a Layer 3 switch could perform routing faster, it is significantly more expensive and unnecessary for a network of this size. The router provides sufficient routing capability while keeping costs low.

Switches were deployed in each section to:

- Reduce cable congestion
- Improve fault isolation
- Simplify maintenance

Labour was assumed to be carried out by a **network technician** rather than a network engineer. This is because the network does not require advanced design complexity that would justify higher labour costs.

Wireless connectivity was implemented using **WRT300N devices configured as access points**. These devices were chosen because they are affordable, easy to configure, and sufficient for the simulation environment.

A contingency of approximately **20%** was assumed to account for unexpected costs or future upgrades.

Remote connectivity was simulated using NAT, allowing internal users to access external networks through a shared connection.

Enterprise-grade solutions such as Layer 3 switches and wireless controllers were considered but not selected due to cost constraints and unnecessary complexity for the network size.

Remote Connectivity & Virtual Office

A VPN was set up using Cisco AnyConnect, which is very secure and works well with Cisco hardware. We looked at using OpenVPN, but preferred Cisco AnyConnect due to better support.

Implications of this for security:

• The tunnels ensure unauthorized interception cannot occur.

• ACL's block movement between VLAN's.

• BYOD's will be placed in a limited-access, secure VLAN.

Virtual office environment:

We have allowed for the use of collaboration tools like Microsoft Teams and Google Workspace so that communication and file sharing can continue, and so remote meetings can take place if we have a fully virtualized office space.

Cost and Budget Considerations

Total Cost Estimate: R95,000- R100,000 is the approximate cost to establish this network at today's market prices for Cisco hardware and inexpensive wireless access points. This amount exceeds the original estimation of R52,800 because more switches and access points are needed than originally estimated for the physical layout and to adequately cover all offices, the reception, the kitchen (IoT VLAN), meeting room, and large open space with over 100 wired access points.

Hardware choices and rationale

Cisco 2960 Switches (12 units, R4000 ea.). Were selected due to enterprise robustness and scalability, along with support for VLANs. Cheaper alternatives were initially considered but dismissed because they would fail more easily and have a shorter lifespan.

Central Router (Cisco ISR, R6000). Was selected as opposed to a Layer 3 switch due to a balance of routing capacity and cost.

Wireless Access points (8 x WRT300N units, R1000 ea.). These were selected to cover all the rooms in the office. High-end wireless controllers were deemed unnecessarily expensive for this network.

Cabling & Accessories(R10000). Include structured cabling, patch panels, racks, etc. To keep cabling neat and expandable in the future.

Labour and Contingency

Labour cost is estimated at R7500 for a network technician. The task should only require approximately 25 hours at R300 per hour, as no expert-level engineering will be required.

Contingency. A 20% contingency is included in the budget of R15900 for unforeseen circumstances, possible replacement hardware, or future expansion.

Hardware Costs

Item	Quantity	Estimated Cost (per unit)	Total
Cisco Switch (2960 series)	12	R4,000	R48,000
Central Router (Cisco ISR)	1	R6,000	R6,000
Wireless Access Points (WRT300N)	8	R1,000	R8,000
Cabling & Accessories	-	-	R10,000
Subtotal (Hardware)			**R72,000**

Labour Costs

Role	Estimated Hours	Rate	Total
Network Technician	25 hrs	R300/hr	R7,500

Total Estimated Cost:

Hardware: R72000

Labour: R7500

Contingency: R15900

Total: R95,400

Justification

This budget balances affordability and robustness. Cisco switches and routers are more expensive than alternative products, but they will last longer and offer more stability. Cheaper wireless access points have been chosen, which can perform all necessary functions at a much lower cost. The contingency has been estimated at 20% in order to ensure that costs remain under the R100,000 maximum, and that future changes or problems do not exceed budget. **Remote Access and Virtual Connectivity**

Remote access to the network can be achieved through VPN (Virtual Private Network) solutions. VPN software such as OpenVPN or Cisco AnyConnect can be used to securely connect remote users to the internal network.

Security Implications

Remote access introduces risks such as:

- Unauthorised access
- Lateral movement within the network

These risks can be mitigated through:

- Strong authentication (e.g., passwords and multi-factor authentication)
- Network segmentation (already implemented via VLANs)

Bring Your Own Device (BYOD)

Users connecting personal devices must be restricted to limited network access. This can be achieved by:

- Placing such devices in a restricted VLAN
- Limiting access using ACLs

Virtual Workspace

Collaboration tools such as Microsoft Teams or Google Workspace can be used to enable communication and file sharing among employees, supporting remote work environments.

Implementation and Design Justification

Network Addressing and Subnetting

A private Class C addressing scheme was used, with each VLAN assigned a **/24 subnet** (255.255.255.0).

Example:

- Network: 192.168.10.0/24
- Usable range: 192.168.10.1 – 192.168.10.254
- Broadcast: 192.168.10.255

Why /24 was chosen:

Although smaller subnets such as /29 or /28 could have been used (especially for areas like reception with only two devices), a /24 subnet was selected for the following reasons:

- **Simplicity**: Using the same subnet size across all VLANs reduces configuration complexity and the risk of errors
- **Scalability**: Future expansion can occur without redesigning the network
- **Consistency**: Makes troubleshooting easier, as all networks follow the same structure

Trade-off

The main disadvantage of /24 is unused IP addresses. However, this was considered acceptable in favour of simplicity and future growth.

3.2 VLAN Segmentation

Each department (offices, kitchen, meeting room, etc.) was assigned its own VLAN.

This was done to:

- Reduce broadcast traffic
- Improve performance
- Enhance security by isolating departments

However, VLANs only provide **Layer 2 isolation**. Once inter-VLAN routing is enabled, devices can still communicate across VLANs.

3.3 Use of ACLs for Isolation

To enforce full isolation, Access Control Lists (ACLs) were implemented on router sub-interfaces.

Why ACLs were necessary

- Inter-VLAN routing allows communication by default.
- This violates the requirement that departments must be isolated.
- ACLs were required to explicitly block this communication.

How ACLs were applied

ACLs were applied on router sub-interfaces (e.g., g0/0.10, g0/0.20) because router-on-a-stick was used. Applying ACLs at this level ensures traffic is filtered before being routed.

Result:

- Department-to-department communication is blocked
- Access to shared resources (printers, internet) is still allowed

3.4 Office Design

Each office area was equipped with its own access switch.

Why were switches placed per office:

- Reduces cable length and clutter
- Improves maintenance (faults are isolated to one office)
- Simplifies expansion (new devices can be added locally)

Wireless placement

Wireless access points were placed centrally rather than per office to reduce hardware cost while still providing adequate coverage.

3.5 Reception Design Decision

The reception was grouped into the same VLAN as the offices.

Why was this decision made

- Reception staff operate at the same trust level as office staff.
- They require access to similar internal resources.
- Separating them would increase configuration complexity without a significant benefit.

Subnet consideration

Although reception only has two devices, a smaller subnet was not used because:

- It would complicate the addressing scheme
- It provides no real advantage in this context

3.6 Kitchen (IoT) Network Design

Kitchen devices were placed in a separate VLAN.

Why:

- IoT devices are generally less secure
- They pose a higher risk to the network
- Isolation prevents them from accessing sensitive systems

3.7 Machine Room and Core Services

The machine room contains the central router responsible for:

- Inter-VLAN routing
- DHCP
- NAT
- ACL enforcement

Why was DHCP implemented on the router

- Reduces need for additional servers.
- Ensures availability (router is always active).
- Simplifies configuration.

Alternative

A dedicated server could have been used, but this would increase cost and complexity without a significant benefit for this network size.

3.8 Wireless Network Design

Wireless routers (WRT300N) were configured as access points by:

- Disabling DHCP
- Assigning static IP addresses
- Connecting via LAN ports

All access points are used:

- Same SSID
- Same security settings

Why

- Allows seamless roaming
- Simplifies user experience

3.9 Internet Connectivity (NAT)

Network Address Translation (NAT) was implemented to allow all internal devices to access the internet through a single external IP.

Why NAT was used

- Conserves public IP addresses
- Provides basic security (internal IPs are hidden)
- Meets the requirement of shared internet access

3.10 Maintenance Considerations

The network was designed to simplify maintenance:

- Distributed switches → faults are isolated
- VLAN segmentation → easier troubleshooting
- Centralised router → easier management

Example

If a switch fails:

- Only that section is affected
- Entire network remains operational

3.11 Assumptions

- Device numbers will grow over time
- All departments require internet access
- Wireless connectivity is required in user areas
- IoT devices present security risks

Discussion

Maintenance Considerations

The access layer switches are most likely to require frequent maintenance, as they are directly connected to end-user devices and are more prone to physical issues such as cable faults.

To improve maintainability:

- Switches should be placed in accessible and secure locations
- Proper cable management should be implemented
- Devices should be clearly labelled

Virtual Office Considerations

If the organisation transitions to a fully virtual environment, the core components of the network would remain relevant, including:

- The central router
- Security configurations (ACLs, NAT)
- Remote access infrastructure

Physical access switches would become less critical, as users would connect remotely rather than from on-site locations.

The network successfully meets all requirements:

- Section isolation
- Internet connectivity
- Printer access
- Wireless connectivity

Strengths

- Scalable design
- Strong security via VLANs and ACLs
- Clear structure

Limitations

- A single router creates a potential single point of failure

Improvement

- Redundancy (backup router) could be added

Network Evaluation

Advantages:

- Hierarchical (easy to scale)
- Isolation (VLANs and ACLs)
- Cheaper hardware options.

Disadvantages:

Single point of failure (due to only one router).

Maintenance:

- Access switches are predicted to require most of the maintenance, due to the direct connection to the users.
- Good cable management and labelling of devices were advised to prevent future problems.

Transition to Virtual:

- If the business switches over to a completely virtual network, the current hardware (router, ACLs, NAT, and VPN server) can be reused
- The access switches would, however, become unnecessary.

Conclusion

Project Management and Collaboration

The project was completed without relying on frequent face-to-face meetings, requiring effective communication and coordination among group members.

Advantages

- Flexible collaboration
- Ability to work independently
- Individual progress was easy
- Used GitHub effectively for version control.

Disadvantages

- Delays in communication
- Difficulty in synchronising configurations

Technical Challenges

Several technical challenges were encountered, including:

- VLAN misconfigurations
- Native VLAN mismatch errors
- ACL errors during isolation
- Wireless DHCP conflict

These challenges were resolved through testing, debugging, and iterative improvements.

Lessons Learned:

This project reinforced the importance of:

- Proper planning before implementation
- Understanding the purpose of each configuration
- Testing each component thoroughly
- Justifying design decisions rather than only implementing them

The project successfully demonstrated the design and implementation of a functional and secure network.

Key concepts applied:

- VLAN segmentation
- ACL security
- DHCP automation
- NAT for internet access

The project also highlighted the importance of:

- Planning
- Testing
- Justifying design decisions

Appendix A: Key Configurations

VLANs:

- vlan 10
- vlan 20
- vlan 30
- vlan 40
- vlan 50
- vlan 60
- vlan 70

Explanation:

VLANs (Virtual Local Area Networks) were created to logically separate different departments into isolated broadcast domains. This reduces unnecessary network traffic and improves security by preventing direct communication between departments at Layer 2.

Access Port Configuration

```
interface fa0/x
switchport mode access
switchport access vlan X
```

Explanation:

Access ports are used to connect end devices such as PCs, printers, and IoT devices to the network. Each port is assigned to a specific VLAN, ensuring that devices are placed in the correct network segment and remain isolated from other departments.

Trunk Configuration

- interface fa0/x
- switchport mode trunk

Explanation:

Trunk ports are used to carry traffic from multiple VLANs between switches or between a switch and a router. This allows a single physical link to transport multiple VLANs, which is essential for inter-VLAN communication and efficient network design.

Router-on-a-Stick Configuration

- interface g0/0.10
- encapsulation dot1Q 10
- ip address 192.168.10.1 255.255.255.0

Explanation:

Router-on-a-stick is used to enable inter-VLAN routing using a single physical router interface. Each subinterface represents a VLAN and is assigned a unique IP address, which acts as the default gateway for devices in that VLAN.

DHCP Configuration

- ip dhcp pool OFFICE
- network 192.168.10.0 255.255.255.0
- default-router 192.168.10.1

Explanation:

DHCP (Dynamic Host Configuration Protocol) was used to automatically assign IP addresses to devices on the network. This eliminates the need for manual configuration and reduces the risk of IP conflicts. The default router specifies the gateway that devices will use to communicate outside their subnet.

NAT Configuration

ip nat inside source list 1 interface g0/1 overload

Explanation:

Network Address Translation (NAT) allows multiple internal devices to share a single public IP address when accessing external networks. The “overload” option enables Port Address Translation (PAT), which maps multiple private IP addresses to one public IP using different ports.

ACL Configuration

- ip access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
- access-list 100 permit ip any any

Explanation:

Access Control Lists (ACLs) were used to control traffic between VLANs. Specific deny rules block communication between departments, while the final “permit ip any any” ensures that allowed traffic (such as internet access and printer communication) is not unintentionally blocked.

Wireless Access Point Configuration (WRT300N)

Configuration Steps:

- DHCP disabled
- Static IP assigned (e.g., 192.168.70.X)
- Connected via LAN port
- SSID configured (e.g., TECH_WIFI)
- WPA2 security enabled

Explanation:

Wireless routers were configured as access points to provide Wi-Fi connectivity without performing routing functions. Disabling DHCP ensures that all devices receive IP addresses from the main router, maintaining consistency across the network. Using the same SSID and security settings allows seamless roaming between access points.

Appendix B: Testing Results:

These tests confirm that VLAN isolation, ACL enforcement, and NAT functionality are operating correctly.

Test	Result	Explanation
Office to Office	pings	Same VLAN
Office to Kitchen	Does not ping	Blocked by ACL
Office to Printer	pings	Allowed
Wi-Fi to Internet	pings	NAT working
Wi-Fi to Office	Does not ping	Isolation working